

under that Protocol Vendor ID (lower 16 bits), similarly to how message [Protocol ID](#) and Protocol Vendor ID are encoded in the [Protocol Header](#).

The following rules apply to the encoding of the [ProtocolCode](#) protocol-specific field:

- [ProtocolCode](#) value 0x0000 SHALL be reserved for use as success placeholder when either a [GeneralCode](#) of *SUCCESS* (0) or *CONTINUE* (10) are present.
- [ProtocolCode](#) value 0xFFFF SHALL be reserved to indicate that no additional protocol-specific status code is available.
- When the [GeneralCode](#) is *FAILURE*, the [ProtocolCode](#) value of 0xFFFF SHOULD NOT be used, since the conveyance of specific error codes assists in troubleshooting.
- [ProtocolCode](#) values 0x0001 through 0xFFFE SHALL be used to convey protocol-specific status indications.

Since protocol-specific status reports are meant to convey more information than generic codes, it is RECOMMENDED to always use a specific [ProtocolCode](#) value, rather than 0xFFFF, unless there are no additional details to convey.

D.3.3. Protocol-specific data ([ProtocolData](#))

The [ProtocolData](#) portion of the StatusReport message is composed of all data beyond the [ProtocolCode](#) field. If a StatusReport message of size *N* octets is received, the first 8 octets of payload encode the [GeneralCode](#), [ProtocolId](#) and [ProtocolCode](#), while the remaining *N* - 8 bytes represent the protocol-specific [ProtocolData](#).

Encoding of the [ProtocolData](#) portion of the payload depends on the [ProtocolId](#) and potentially [ProtocolCode](#). To decode this data, the [ProtocolId](#) has to be examined and decoding SHALL be done according to that protocol specification. For example:

- A vendor-specific protocol would encode additional custom error metadata in the [ProtocolData](#).
- The [Bulk transfer \(BDX\)](#) protocol does not require additional error information and will always have [ProtocolData](#) empty.

D.4. Presenting [StatusReport](#) messages in protocol specifications

In order to simplify referring to StatusReport messages, the following mnemonic encoding will be used in the descriptive text for a given protocol.

References to [StatusReport](#) messages take one of the following forms:

- No [ProtocolData](#) present:
 - [StatusReport](#)([GeneralCode](#): <value>, [ProtocolId](#): <value>, [ProtocolCode](#): <value>)
 - Example 1: [StatusReport](#)([GeneralCode](#): *FAILURE*, [ProtocolId](#): *BDX*, [ProtocolCode](#): *START_OFFSET_NOT_SUPPORTED*)
 - Encodes as: 01 00 02 00 00 00 52 00